

NIST SP 800-22 Rev. 1a Evaluation of the Authenticated-TPE Method

Scope and interpretation

This report presents separate statistical diagnostics for authenticated-TPE cryptographic components and for the image outputs. NIST SP 800-22 is a statistical test suite, not a security proof. A test outcome does not establish security or insecurity by itself.

Method and implementation configuration

The implementation processes 512x512 RGB uint8 arrays in non-overlapping 32x32 blocks. It uses 256 blocks per channel, 512 adjacent pixel pairs per block, groups four ordered blocks, 64 groups per channel and 192 groups per RGB image. Authentication uses HMAC-SHA256; grouping uses the implemented HMAC-DRBG; transforms are Step 1 pair-sum-preserving mapping, Step 2 image-derived keyed smaller-pixel re-encryption, and reversible contrast mapping (RCM) for tag embedding. Verification and exact recovery occur before plaintext release.

The cryptographic-component categories test outputs used by the implementation. The Authentication-tag stream is raw 256-bit HMAC output, tested separately using independently generated keys and ImageIDs. The image-output categories are explicitly diagnostic and preserve the method's image-dependent structure.

NIST software and settings

- Software: official NIST Statistical Test Suite 2.1.2, selected from the NIST SP 800-22 documentation and software page.
- Tests: all 15 STS tests; alpha = 0.01; default parameters (block frequency 128, non-overlapping templates m=9, overlapping template m=9, approximate entropy m=10, serial m=16, linear complexity M=500).
- Sample: 100 streams for each category; each stream is exactly 1,000,000 bits; 600 streams total across six categories.
- Input: NIST binary mode. The runner confirmed 100 BITSREAD = 1000000 records per category. Each 125,000-byte stream is consumed MSB-first, matching the NIST convertToBits implementation.
- NIST's native individual p-value output is six decimal places. The CSV preserves that output; a displayed 0.000000 is rounded output, not a missing p-value. No synthetic precision is added.
- Deterministic component fixture bytes use SHA-256(domain_label || 0x00 || uint64_be(index)), truncated only where a 16-byte ImageID is needed. The exact domain labels, per-stream key/nonce digests, and fixture indexes are recorded in stream_manifest.csv and component_contexts.csv.

Exact stream-generation procedure

- cryptographic_hmac_drbg: 100 independent streams. Each uses a unique 32-byte deterministic test entropy input and 16-byte nonce, then the authenticated-TPE HMAC_DRBG(SHA-256) implementation with personalization 'group-perm'. Each stream concatenates 31,250 successive 4-byte Generate outputs (1,000,000 bits). This extends the same implementation beyond the 255 Fisher-Yates draws typically consumed per channel; no value is repeated or padded.
- cryptographic_pair_shift_words: 100 independent UserKey/ImageID fixtures. Each stream contains the first 31,250 distinct valid (channel, block_id, pair_index) HMAC inputs, ordered channel 0..2, block 0..255, pair 0..511. Each output is the exact first 32 digest bits used by pair_shift before the implementation's class-size modulo. This tests the raw HMAC-derived word, not a fabricated image and not post-modulo shift values.
- cryptographic_block_r1_values: Each 1,000,000-bit sequence is the exact two-bit block_r1 output across independent UserKey contexts in context/channel/block order. A single context yields only 768 values, so 652 independent key contexts are used per NIST stream; the last context contributes only the 32 values needed for the exact prefix. Each r1 is encoded as two MSB-first bits. Keys are not reused and no values are repeated or padded.
- Authentication-tag stream: 100 separate NIST bitstreams are assembled from raw 256-bit HMAC-SHA256 group tags generated by the implementation _group_mac. Each stream uses 21 independent UserKey/ImageID contexts; all 2,100 pairs are unique. Six fixed Step-1 public UCT reference images supply authenticated group data round-robin. Tags are concatenated in context/channel/group order (4,032 tags generated per sequence) and truncated at exactly 1,000,000 bits, within the final tag bytes. This is an isolated tag-component diagnostic, not 100 full image-protection runs.
- image_output_step2: the first 1,000,000 bits of row-major C-order RGB uint8 bytes from the Step-2 image-derived keyed intermediate, before authentication-tag embedding.
- image_output_final_marked: the first 1,000,000 bits of row-major C-order RGB uint8 bytes from the final RCM-marked authenticated image.

Image and key fixture selection

The image diagnostic uses the six available public UCT images in round-robin order: airplane, baboon, couple, girl, lena, peppers. Images are converted to RGB and resized to 512x512 only when required, using Pillow's default resize behavior to match the existing runner. For stream i, UserKey is SHA-256(auth-tpe-nist-key-v1 || 00 || uint32_be(i)) and ImageID is the first 16 bytes of SHA-256(auth-tpe-nist-image-id-v1 || 00 || uint32_be(i)). The paired Step-2 and marked outputs for one stream use the same test credentials. These are reproducibility fixtures, not production secrets. The exact eight CelebA-HQ source files were unavailable; no CelebA-HQ data is claimed or fabricated.

The four cryptographic-component fixture families use domain-separated deterministic public bytes: HMAC-DRBG entropy/nonce, pair-shift key/ImageID, block-r1 keys, and authentication-tag key/ImageID. The tag category uses 2,100 unique UserKey/ImageID pairs across the 100 streams. These fixtures support repeatability and are public; they do not test entropy sources or secret-key security.

Byte-to-bit serialization

Cryptographic byte outputs are recorded in generation order. For the image categories, the tested object is row-major C-order RGB uint8 bytes. The first 125,000 bytes are the exact 1,000,000-bit prefix. All inputs are passed as packed binary; NIST reads every byte from its most significant bit to its least significant bit. No LSB-first or full-image variant was run, and no image-output padding was used.

Results overview

Cryptographic-component evaluation

Category	Pass	Fail	N/A
cryptographic_hmac_drbg	17328	172	1300
cryptographic_pair_shift_words	17537	171	1092
cryptographic_block_r1_values	17698	192	910
Authentication-tag stream	17631	181	988

Image-output statistical diagnostic

Category	Pass	Fail	N/A	Exact recovery
image_output_step2	1838	14362	2600	100/100
image_output_final_marked	1938	14262	2600	100/100

The image outputs retain source-image and RCM/tag structure by design. Their image-output tests are diagnostics and should not be expected to behave like an ideal random-number stream. Low p-values in those categories do not by themselves establish a cryptographic implementation defect. The cryptographic component results assess pseudorandom outputs separately, but they also do not constitute a security proof.

Full test/component tables

The tables below give, for every NIST test/component, the number of valid streams, first-level pass/fail/N/A counts, native NIST uniformity p-value, and pass proportion. The uniformity p-value is a second-level value over first-level p-values and is not an individual stream p-value.

cryptographic_hmac_drbg

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
Frequency	single	100	99	1	0	0.304126	99/100
BlockFrequency	single	100	99	1	0	0.153763	99/100
CumulativeSums	forward	100	99	1	0	0.051942	99/100
CumulativeSums	reverse	100	99	1	0	0.289667	99/100
Runs	single	100	98	2	0	0.181557	98/100
LongestRun	single	100	97	3	0	0.971699	97/100
Rank	single	100	100	0	0	0.883171	100/100
FFT	single	100	100	0	0	0.224821	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 0 1	100	100	0	0	0.213309	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 1 1	100	100	0	0	0.994250	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 0 1	100	99	1	0	0.534146	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 1 1	100	98	2	0	0.319084	98/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 0 1	100	100	0	0	0.045675	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 1 1	100	97	3	0	0.987896	97/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 0 1	100	99	1	0	0.964295	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 1 1	100	100	0	0	0.145326	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	99	1	0	0.153763	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	97	3	0	0.236810	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	99	1	0	0.595549	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	100	0	0	0.383827	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	99	1	0	0.924076	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	100	0	0	0.366918	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	99	1	0	0.739918	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	98	2	0	0.678686	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	99	1	0	0.534146	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 0	100	99	1	0	0.055361	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	99	1	0	0.987896	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 0	100	100	0	0	0.042808	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	100	0	0	0.678686	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 0	100	98	2	0	0.514124	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	100	0	0	0.289667	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	98	2	0	0.129620	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 0	100	98	2	0	0.262249	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	98	2	0	0.494392	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 0	100	99	1	0	0.897763	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	99	1	0	0.883171	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 0	100	100	0	0	0.249284	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 0	100	100	0	0	0.739918	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 0 1	100	99	1	0	0.991468	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 0	100	100	0	0	0.213309	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 1	100	99	1	0	0.115387	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 0 0	100	100	0	0	0.319084	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 0 1	100	99	1	0	0.851383	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 0	100	98	2	0	0.779188	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 0 0	100	99	1	0	0.759756	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 0 1	100	100	0	0	0.455937	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 0	100	100	0	0	0.224821	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 1	100	98	2	0	0.699313	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 0 0	100	99	1	0	0.115387	99/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 1 1	100	98	2	0	0.637119	98/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 0 1	100	97	3	0	0.616305	97/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 1 1	100	98	2	0	0.289667	98/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	99	1	0	0.514124	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 0 1	100	100	0	0	0.040108	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 1 1	100	100	0	0	0.058984	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 0 1	100	100	0	0	0.798139	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 1 1	100	100	0	0	0.699313	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 0 1 1	100	99	1	0	0.867692	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 0 1	100	100	0	0	0.867692	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 1 1	100	100	0	0	0.249284	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1	100	99	1	0	0.191687	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1	100	100	0	0	0.657933	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 0 1 1	100	100	0	0	0.554420	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1 1	100	100	0	0	0.275709	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 0 1 1	100	99	1	0	0.678686	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1 1	100	100	0	0	0.437274	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 0 1 1	100	98	2	0	0.122325	98/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1 1	100	99	1	0	0.437274	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 0 1 1	100	100	0	0	0.048716	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1 1	100	100	0	0	0.816537	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1 1	100	97	3	0	0.897763	97/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1 1	100	100	0	0	0.739918	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 0 1 1	100	100	0	0	0.851383	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1 1	100	100	0	0	0.319084	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 0 1 1	100	98	2	0	0.334538	98/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1 1	100	99	1	0	0.955835	99/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 0 1 1 1	100	97	3	0	0.334538	97/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 1 1 1 1	100	99	1	0	0.897763	99/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 0 1 1 1	100	100	0	0	0.851383	100/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 1 1 1 1	100	99	1	0	0.494392	99/100
NonOverlappingTemplate	m=9 template=0 1 1 1 0 1 1 1 1	100	99	1	0	0.249284	99/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 1 1 1 1	100	100	0	0	0.779188	100/100
NonOverlappingTemplate	m=9 template=1 0 0 0 0 0 0 0 0	100	100	0	0	0.213309	100/100
NonOverlappingTemplate	m=9 template=1 0 0 0 1 0 0 0 0	100	99	1	0	0.739918	99/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 0 0 0 0	100	97	3	0	0.304126	97/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 1 0 0 0	100	98	2	0	0.534146	98/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 0 0 0 0	100	98	2	0	0.971699	98/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 1 0 0 0	100	100	0	0	0.554420	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 0 0 0	100	98	2	0	0.534146	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 1 0 0	100	100	0	0	0.474986	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 0 0 0	100	98	2	0	0.289667	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 1 0 0	100	98	2	0	0.236810	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0 0	100	99	1	0	0.437274	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 1 0 0	100	99	1	0	0.739918	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0 0	100	99	1	0	0.202268	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 1 0 0	100	98	2	0	0.678686	98/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 0 0 0	100	99	1	0	0.334538	99/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 1 0 0	100	100	0	0	0.224821	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 0 0 0	100	100	0	0	0.319084	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 1 0 0	100	100	0	0	0.678686	100/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 0 0 0	100	100	0	0	0.153763	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 1 0 0	100	100	0	0	0.102526	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 0 0 0	100	96	4	0	0.678686	96/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 1 0 0	100	97	3	0	0.162606	97/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 0 0	100	100	0	0	0.514124	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 1 0 0	100	99	1	0	0.971699	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	99	1	0	0.366918	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	100	0	0	0.122325	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 1 0	100	100	0	0	0.249284	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 0 0	100	99	1	0	0.455937	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 1 0	100	100	0	0	0.883171	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 1 0 0	100	98	2	0	0.924076	98/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 0 0	100	99	1	0	0.419021	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 1 0	100	100	0	0	0.419021	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 0 0	100	100	0	0	0.494392	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 1 0	100	97	3	0	0.554420	97/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 1 0 0	100	99	1	0	0.236810	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 0 0	100	99	1	0	0.075719	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 1 0	100	98	2	0	0.080519	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 1 0 0	100	98	2	0	0.401199	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 0 0	100	99	1	0	0.023545	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 1 0	100	98	2	0	0.042808	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 1 0 0	100	100	0	0	0.035174	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 0 0	100	99	1	0	0.019188	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 1 0	100	99	1	0	0.616305	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 1 0 0	100	99	1	0	0.334538	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 0 0	100	100	0	0	0.051942	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 1 0	100	99	1	0	0.334538	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 0 0	100	99	1	0	0.474986	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 1 0	100	99	1	0	0.350485	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 0 0	100	100	0	0	0.911413	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 1 0	100	98	2	0	0.474986	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 1 0 0	100	98	2	0	0.319084	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 0 0	100	99	1	0	0.040108	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 1 0	100	100	0	0	0.162606	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 0 0	100	98	2	0	0.096578	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 1 0	100	99	1	0	0.719747	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 0 0	100	100	0	0	0.080519	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 1 0	100	99	1	0	0.289667	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 1 0 0	100	97	3	0	0.275709	97/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 0 0	100	99	1	0	0.009535	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 1 0	100	98	2	0	0.924076	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 0 0	100	99	1	0	0.066882	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 1 0	100	100	0	0	0.058984	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 0 0	100	98	2	0	0.102526	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 1 0	100	100	0	0	0.514124	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0 0	100	98	2	0	0.996335	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 1 0	100	98	2	0	0.115387	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0 0	100	98	2	0	0.834308	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 1 0	100	99	1	0	0.595549	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0 0	100	98	2	0	0.657933	98/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0	100	100	0	0	0.000474	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0	100	100	0	0	0.383827	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0	100	99	1	0	0.739918	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 0	100	99	1	0	0.090936	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 0	100	100	0	0	0.816537	100/100
OverlappingTemplate	single	100	96	4	0	0.911413	96/100
Universal	single	100	97	3	0	0.514124	97/100
ApproximateEntropy	single	100	99	1	0	0.129620	99/100
RandomExcursions	state=-4	50	50	0	50	0.699313	50/50
RandomExcursions	state=-3	50	49	1	50	0.935716	49/50
RandomExcursions	state=-2	50	50	0	50	0.262249	50/50
RandomExcursions	state=-1	50	50	0	50	0.971699	50/50
RandomExcursions	state=1	50	50	0	50	0.137282	50/50
RandomExcursions	state=2	50	50	0	50	0.383827	50/50
RandomExcursions	state=3	50	50	0	50	0.494392	50/50
RandomExcursions	state=4	50	50	0	50	0.096578	50/50
RandomExcursionsVariant	state=-9	50	49	1	50	0.236810	49/50
RandomExcursionsVariant	state=-8	50	49	1	50	0.108791	49/50
RandomExcursionsVariant	state=-7	50	49	1	50	0.013569	49/50
RandomExcursionsVariant	state=-6	50	49	1	50	0.350485	49/50
RandomExcursionsVariant	state=-5	50	49	1	50	0.289667	49/50
RandomExcursionsVariant	state=-4	50	49	1	50	0.494392	49/50
RandomExcursionsVariant	state=-3	50	49	1	50	0.616305	49/50
RandomExcursionsVariant	state=-2	50	50	0	50	0.699313	50/50
RandomExcursionsVariant	state=-1	50	49	1	50	0.236810	49/50
RandomExcursionsVariant	state=1	50	49	1	50	0.494392	49/50
RandomExcursionsVariant	state=2	50	50	0	50	0.262249	50/50
RandomExcursionsVariant	state=3	50	50	0	50	0.699313	50/50
RandomExcursionsVariant	state=4	50	50	0	50	0.213309	50/50
RandomExcursionsVariant	state=5	50	50	0	50	0.383827	50/50
RandomExcursionsVariant	state=6	50	50	0	50	0.191687	50/50
RandomExcursionsVariant	state=7	50	50	0	50	0.699313	50/50
RandomExcursionsVariant	state=8	50	50	0	50	0.494392	50/50
RandomExcursionsVariant	state=9	50	50	0	50	0.153763	50/50
Serial	p_value1 (delta_1)	100	100	0	0	0.224821	100/100
Serial	p_value2 (delta_2)	100	100	0	0	0.080519	100/100
LinearComplexity	single	100	99	1	0	0.401199	99/100

cryptographic_pair_shift_words

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
Frequency	single	100	99	1	0	0.191687	99/100
BlockFrequency	single	100	98	2	0	0.191687	98/100
CumulativeSums	forward	100	98	2	0	0.851383	98/100
CumulativeSums	reverse	100	98	2	0	0.946308	98/100
Runs	single	100	99	1	0	0.883171	99/100
LongestRun	single	100	99	1	0	0.249284	99/100
Rank	single	100	100	0	0	0.514124	100/100
FFT	single	100	99	1	0	0.122325	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 0 1	100	98	2	0	0.494392	98/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 1 1	100	99	1	0	0.574903	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 0 1	100	99	1	0	0.534146	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 1 1	100	99	1	0	0.236810	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 0 1	100	99	1	0	0.739918	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 1 1	100	99	1	0	0.637119	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 0 1	100	98	2	0	0.798139	98/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 1 1	100	98	2	0	0.137282	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	97	3	0	0.494392	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	98	2	0	0.289667	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	98	2	0	0.071177	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	100	0	0	0.437274	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	100	0	0	0.911413	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	99	1	0	0.055361	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	99	1	0	0.334538	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	97	3	0	0.798139	97/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 1	100	100	0	0	0.162606	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 0 1	100	100	0	0	0.162606	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 1	100	97	3	0	0.616305	97/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 0 1	100	99	1	0	0.319084	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 1	100	98	2	0	0.262249	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 0 1	100	98	2	0	0.419021	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 1	100	99	1	0	0.699313	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 1	100	100	0	0	0.202268	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1	100	100	0	0	0.657933	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 1	100	99	1	0	0.719747	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 0 1	100	97	3	0	0.455937	97/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 1	100	99	1	0	0.657933	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 0 1	100	100	0	0	0.719747	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 1	100	96	4	0	0.437274	96/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1	100	98	2	0	0.021999	98/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 0 1	100	98	2	0	0.978072	98/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1	100	99	1	0	0.304126	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1	100	98	2	0	0.678686	98/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 0 1	100	98	2	0	0.115387	98/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1	100	100	0	0	0.883171	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1	100	99	1	0	0.014550	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1	100	99	1	0	0.350485	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1	100	99	1	0	0.834308	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1	100	100	0	0	0.366918	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 0 1	100	99	1	0	0.249284	99/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 1 1	100	99	1	0	0.401199	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 0 1	100	99	1	0	0.334538	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 1 1	100	100	0	0	0.090936	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	100	0	0	0.319084	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 0 1	100	100	0	0	0.955835	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 1 1	100	100	0	0	0.637119	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 0 1	100	97	3	0	0.275709	97/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 1 1	100	99	1	0	0.554420	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 0 1 1	100	100	0	0	0.657933	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 0 1	100	100	0	0	0.137282	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 1 1	100	99	1	0	0.971699	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1	100	99	1	0	0.162606	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1	100	100	0	0	0.851383	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1	100	99	1	0	0.304126	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1	100	98	2	0	0.883171	98/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1	100	99	1	0	0.236810	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1	100	99	1	0	0.911413	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1	100	99	1	0	0.574903	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1	100	99	1	0	0.071177	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1	100	97	3	0	0.983453	97/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1	100	99	1	0	0.275709	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1	100	100	0	0	0.383827	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1	100	97	3	0	0.759756	97/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 0 1	100	100	0	0	0.534146	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1	100	99	1	0	0.494392	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 0 1	100	99	1	0	0.595549	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1	100	100	0	0	0.851383	100/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 0 1 1	100	98	2	0	0.334538	98/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 1 1 1	100	100	0	0	0.401199	100/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 0 1 1	100	100	0	0	0.851383	100/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 1 1 1	100	100	0	0	0.334538	100/100
NonOverlappingTemplate	m=9 template=0 1 1 1 0 1 1 1	100	97	3	0	0.334538	97/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 1 1 1	100	98	2	0	0.236810	98/100
NonOverlappingTemplate	m=9 template=1 0 0 0 0 0 0 0	100	98	2	0	0.494392	98/100
NonOverlappingTemplate	m=9 template=1 0 0 0 1 0 0 0	100	98	2	0	0.383827	98/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 0 0 0	100	100	0	0	0.897763	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 1 0 0	100	100	0	0	0.202268	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 0 0 0	100	100	0	0	0.383827	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 1 0 0	100	97	3	0	0.514124	97/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 0 0	100	98	2	0	0.935716	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 0 0	100	97	3	0	0.096578	97/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0	100	99	1	0	0.987896	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0	100	98	2	0	0.987896	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0	100	99	1	0	0.419021	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 1 0	100	99	1	0	0.779188	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0	100	97	3	0	0.040108	97/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 1 0	100	100	0	0	0.494392	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 0 0	100	100	0	0	0.987896	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 1 0	100	100	0	0	0.145326	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 0 0	100	98	2	0	0.289667	98/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 1 0	100	98	2	0	0.366918	98/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 0 0 0	100	99	1	0	0.798139	99/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 1 0 0	100	100	0	0	0.455937	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 0 0 0	100	99	1	0	0.514124	99/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 1 0 0	100	99	1	0	0.007694	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 0 0	100	99	1	0	0.474986	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 1 0 0	100	100	0	0	0.883171	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	100	0	0	0.249284	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	100	0	0	0.455937	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 1 0	100	100	0	0	0.678686	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 0 0	100	100	0	0	0.911413	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 1 0	100	100	0	0	0.213309	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 1 0 0	100	100	0	0	0.401199	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 0 0	100	100	0	0	0.867692	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 1 0	100	98	2	0	0.032923	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 0 0	100	98	2	0	0.798139	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 1 0	100	99	1	0	0.366918	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 1 0 0	100	99	1	0	0.851383	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 0 0	100	99	1	0	0.924076	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 1 0	100	98	2	0	0.115387	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 1 0 0	100	99	1	0	0.637119	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 0 0	100	100	0	0	0.045675	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 1 0	100	100	0	0	0.366918	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 1 0 0	100	100	0	0	0.657933	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 0 0	100	99	1	0	0.366918	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 1 0	100	99	1	0	0.494392	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 1 0 0	100	100	0	0	0.304126	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 0 0	100	98	2	0	0.816537	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 1 0	100	100	0	0	0.739918	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 0 0	100	100	0	0	0.401199	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 1 0	100	98	2	0	0.883171	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 0 0	100	98	2	0	0.719747	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 1 0	100	99	1	0	0.350485	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 1 0 0	100	98	2	0	0.494392	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 0 0	100	100	0	0	0.699313	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 1 0	100	99	1	0	0.534146	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 0 0	100	99	1	0	0.897763	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 1 0	100	98	2	0	0.437274	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 0 0	100	100	0	0	0.383827	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 1 0	100	99	1	0	0.574903	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 1 0 0	100	99	1	0	0.010988	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 0 0	100	99	1	0	0.062821	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 1 0	100	100	0	0	0.236810	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 0 0	100	100	0	0	0.350485	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 1 0	100	100	0	0	0.955835	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 0 0	100	99	1	0	0.035174	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 1 0	100	98	2	0	0.574903	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0 0	100	99	1	0	0.616305	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 1 0	100	98	2	0	0.419021	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0 0	100	100	0	0	0.657933	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 1 0	100	97	3	0	0.883171	97/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0 0	100	99	1	0	0.096578	99/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0	100	99	1	0	0.122325	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0	100	99	1	0	0.191687	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0	100	99	1	0	0.494392	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 0	100	99	1	0	0.474986	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 0	100	98	2	0	0.236810	98/100
OverlappingTemplate	single	100	100	0	0	0.145326	100/100
Universal	single	100	100	0	0	0.289667	100/100
ApproximateEntropy	single	100	100	0	0	0.020548	100/100
RandomExcursions	state=-4	58	56	2	42	0.066882	56/58
RandomExcursions	state=-3	58	57	1	42	0.657933	57/58
RandomExcursions	state=-2	58	58	0	42	0.001509	58/58
RandomExcursions	state=-1	58	58	0	42	0.383827	58/58
RandomExcursions	state=1	58	58	0	42	0.013569	58/58
RandomExcursions	state=2	58	58	0	42	0.383827	58/58
RandomExcursions	state=3	58	58	0	42	0.236810	58/58
RandomExcursions	state=4	58	58	0	42	0.350485	58/58
RandomExcursionsVariant	state=-9	58	58	0	42	0.739918	58/58
RandomExcursionsVariant	state=-8	58	57	1	42	0.236810	57/58
RandomExcursionsVariant	state=-7	58	58	0	42	0.011791	58/58
RandomExcursionsVariant	state=-6	58	58	0	42	0.816537	58/58
RandomExcursionsVariant	state=-5	58	58	0	42	0.657933	58/58
RandomExcursionsVariant	state=-4	58	58	0	42	0.534146	58/58
RandomExcursionsVariant	state=-3	58	58	0	42	0.236810	58/58
RandomExcursionsVariant	state=-2	58	58	0	42	0.419021	58/58
RandomExcursionsVariant	state=-1	58	58	0	42	0.045675	58/58
RandomExcursionsVariant	state=1	58	58	0	42	0.051942	58/58
RandomExcursionsVariant	state=2	58	58	0	42	0.289667	58/58
RandomExcursionsVariant	state=3	58	58	0	42	0.075719	58/58
RandomExcursionsVariant	state=4	58	58	0	42	0.262249	58/58
RandomExcursionsVariant	state=5	58	58	0	42	0.030806	58/58
RandomExcursionsVariant	state=6	58	58	0	42	0.213309	58/58
RandomExcursionsVariant	state=7	58	58	0	42	0.699313	58/58
RandomExcursionsVariant	state=8	58	58	0	42	0.419021	58/58
RandomExcursionsVariant	state=9	58	58	0	42	0.213309	58/58
Serial	p_value1 (delta_1)	100	100	0	0	0.289667	100/100
Serial	p_value2 (delta_2)	100	100	0	0	0.534146	100/100
LinearComplexity	single	100	100	0	0	0.102526	100/100

cryptographic_block_r1_values

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
Frequency	single	100	99	1	0	0.574903	99/100
BlockFrequency	single	100	98	2	0	0.595549	98/100
CumulativeSums	forward	100	99	1	0	0.005762	99/100
CumulativeSums	reverse	100	99	1	0	0.955835	99/100
Runs	single	100	99	1	0	0.236810	99/100
LongestRun	single	100	99	1	0	0.096578	99/100
Rank	single	100	98	2	0	0.798139	98/100
FFT	single	100	99	1	0	0.213309	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 0 1	100	99	1	0	0.779188	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 1 1	100	99	1	0	0.935716	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 0 1	100	100	0	0	0.946308	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 1 1	100	100	0	0	0.062821	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 0 1	100	98	2	0	0.202268	98/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 1 1	100	98	2	0	0.554420	98/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 0 1	100	100	0	0	0.834308	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 1 1	100	98	2	0	0.897763	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	99	1	0	0.554420	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	99	1	0	0.657933	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	100	0	0	0.834308	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	98	2	0	0.304126	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	99	1	0	0.595549	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	97	3	0	0.851383	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	98	2	0	0.574903	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	99	1	0	0.851383	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	99	1	0	0.883171	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 0	100	100	0	0	0.366918	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	100	0	0	0.554420	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 0	100	100	0	0	0.494392	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	99	1	0	0.494392	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 0	100	99	1	0	0.867692	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	98	2	0	0.867692	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	98	2	0	0.350485	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 0	100	100	0	0	0.289667	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	99	1	0	0.108791	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 0	100	100	0	0	0.946308	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	98	2	0	0.964295	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 0	100	100	0	0	0.181557	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	100	0	0	0.971699	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 0 1	100	100	0	0	0.978072	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 0	100	96	4	0	0.779188	96/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 1	100	100	0	0	0.171867	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 0 1	100	99	1	0	0.048716	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 0	100	98	2	0	0.851383	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 1	100	99	1	0	0.145326	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 0 1	100	100	0	0	0.978072	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 0	100	98	2	0	0.657933	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 1	100	100	0	0	0.224821	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 0 1	100	98	2	0	0.419021	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 0	100	99	1	0	0.554420	99/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 1 1	100	99	1	0	0.924076	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 0 1	100	99	1	0	0.334538	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 1 1	100	100	0	0	0.401199	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	99	1	0	0.419021	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 0 1	100	100	0	0	0.108791	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 1 1	100	100	0	0	0.595549	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 0 1	100	100	0	0	0.851383	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 1 1	100	99	1	0	0.719747	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 0 1 1	100	98	2	0	0.262249	98/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 0 1	100	100	0	0	0.595549	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 1 1	100	97	3	0	0.911413	97/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	98	2	0	0.334538	98/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1 1	100	99	1	0	0.350485	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1 1	100	99	1	0	0.122325	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1 1	100	100	0	0	0.015598	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1 1	100	97	3	0	0.935716	97/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1 1	100	99	1	0	0.637119	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1 1	100	97	3	0	0.236810	97/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1 1	100	97	3	0	0.759756	97/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1 1	100	99	1	0	0.062821	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1 1	100	100	0	0	0.798139	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1 1	100	100	0	0	0.554420	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1 1	100	99	1	0	0.191687	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 0 1 1	100	99	1	0	0.699313	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1 1	100	98	2	0	0.419021	98/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 0 1 1	100	99	1	0	0.574903	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1 1	100	98	2	0	0.616305	98/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 0 1 1 1	100	97	3	0	0.401199	97/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 1 1 1 1	100	98	2	0	0.759756	98/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 0 1 1 1	100	99	1	0	0.678686	99/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 1 1 1 1	100	100	0	0	0.350485	100/100
NonOverlappingTemplate	m=9 template=0 1 1 1 0 1 1 1 1	100	99	1	0	0.935716	99/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 1 1 1 1	100	99	1	0	0.657933	99/100
NonOverlappingTemplate	m=9 template=1 0 0 0 0 0 0 0 0	100	99	1	0	0.779188	99/100
NonOverlappingTemplate	m=9 template=1 0 0 0 1 0 0 0 0	100	100	0	0	0.037566	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 0 0 0 0	100	98	2	0	0.554420	98/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 1 0 0 0	100	100	0	0	0.383827	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 0 0 0 0	100	100	0	0	0.383827	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 1 0 0 0	100	100	0	0	0.171867	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 0 0 0	100	99	1	0	0.249284	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 1 0 0	100	99	1	0	0.058984	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 0 0 0	100	96	4	0	0.816537	96/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 1 0 0	100	99	1	0	0.334538	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0 0	100	99	1	0	0.383827	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 1 0 0	100	96	4	0	0.108791	96/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0 0	100	100	0	0	0.090936	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 1 0 0	100	98	2	0	0.236810	98/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 0 0 0	100	100	0	0	0.319084	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 1 0 0	100	100	0	0	0.595549	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 0 0 0	100	97	3	0	0.719747	97/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 1 0 0	100	99	1	0	0.062821	99/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 0 0 0	100	99	1	0	0.366918	99/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 1 0 0	100	98	2	0	0.554420	98/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 0 0 0	100	100	0	0	0.964295	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 1 0 0	100	100	0	0	0.350485	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 0 0	100	100	0	0	0.971699	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 1 0	100	99	1	0	0.637119	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 1 0 0	100	99	1	0	0.534146	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	99	1	0	0.616305	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 1 0	100	99	1	0	0.924076	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 0 0	100	98	2	0	0.366918	98/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 1 0	100	98	2	0	0.383827	98/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 1 0 0	100	97	3	0	0.350485	97/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 0 0	100	100	0	0	0.616305	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 1 0	100	100	0	0	0.455937	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 0 0	100	100	0	0	0.574903	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 1 0	100	99	1	0	0.719747	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 1 0 0	100	99	1	0	0.213309	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 0 0	100	97	3	0	0.759756	97/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 1 0	100	100	0	0	0.249284	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 1 0 0	100	100	0	0	0.401199	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 0 0	100	99	1	0	0.759756	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 1 0	100	98	2	0	0.236810	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 1 0 0	100	100	0	0	0.739918	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 0 0	100	99	1	0	0.759756	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 1 0	100	97	3	0	0.759756	97/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 1 0 0	100	99	1	0	0.014550	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 0 0	100	98	2	0	0.798139	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 1 0	100	99	1	0	0.946308	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 0 0	100	99	1	0	0.924076	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 1 0	100	99	1	0	0.474986	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 0 0	100	95	5	0	0.437274	95/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 1 0	100	98	2	0	0.202268	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 1 0 0	100	99	1	0	0.657933	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 0 0	100	100	0	0	0.739918	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 1 0	100	100	0	0	0.319084	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 0 0	100	97	3	0	0.834308	97/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 1 0	100	99	1	0	0.534146	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 0 0	100	99	1	0	0.924076	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 1 0	100	99	1	0	0.437274	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 1 0 0	100	98	2	0	0.319084	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 0 0	100	97	3	0	0.474986	97/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 1 0	100	98	2	0	0.319084	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 0 0	100	100	0	0	0.514124	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 1 0	100	99	1	0	0.319084	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 0 0	100	100	0	0	0.401199	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 1 0	100	100	0	0	0.911413	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0 0	100	100	0	0	0.999777	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 1 0	100	100	0	0	0.657933	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0 0	100	99	1	0	0.437274	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 1 0	100	100	0	0	0.236810	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0 0	100	99	1	0	0.657933	99/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0	100	100	0	0	0.401199	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0	100	99	1	0	0.040108	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0	100	100	0	0	0.236810	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 0	100	98	2	0	0.366918	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 0	100	99	1	0	0.719747	99/100
OverlappingTemplate	single	100	100	0	0	0.779188	100/100
Universal	single	100	100	0	0	0.554420	100/100
ApproximateEntropy	single	100	99	1	0	0.759756	99/100
RandomExcursions	state=-4	65	65	0	35	0.204076	65/65
RandomExcursions	state=-3	65	65	0	35	0.242986	65/65
RandomExcursions	state=-2	65	63	2	35	0.264458	63/65
RandomExcursions	state=-1	65	64	1	35	0.922036	64/65
RandomExcursions	state=1	65	64	1	35	0.186566	64/65
RandomExcursions	state=2	65	65	0	35	0.585209	65/65
RandomExcursions	state=3	65	63	2	35	0.585209	63/65
RandomExcursions	state=4	65	65	0	35	0.170294	65/65
RandomExcursionsVariant	state=-9	65	64	1	35	0.551026	64/65
RandomExcursionsVariant	state=-8	65	64	1	35	0.051391	64/65
RandomExcursionsVariant	state=-7	65	64	1	35	0.689019	64/65
RandomExcursionsVariant	state=-6	65	64	1	35	0.551026	64/65
RandomExcursionsVariant	state=-5	65	64	1	35	0.689019	64/65
RandomExcursionsVariant	state=-4	65	64	1	35	0.392456	64/65
RandomExcursionsVariant	state=-3	65	64	1	35	0.170294	64/65
RandomExcursionsVariant	state=-2	65	63	2	35	0.452799	63/65
RandomExcursionsVariant	state=-1	65	65	0	35	0.013411	65/65
RandomExcursionsVariant	state=1	65	65	0	35	0.941144	65/65
RandomExcursionsVariant	state=2	65	65	0	35	0.585209	65/65
RandomExcursionsVariant	state=3	65	65	0	35	0.392456	65/65
RandomExcursionsVariant	state=4	65	65	0	35	0.756476	65/65
RandomExcursionsVariant	state=5	65	65	0	35	0.585209	65/65
RandomExcursionsVariant	state=6	65	65	0	35	0.585209	65/65
RandomExcursionsVariant	state=7	65	65	0	35	0.517442	65/65
RandomExcursionsVariant	state=8	65	65	0	35	0.452799	65/65
RandomExcursionsVariant	state=9	65	65	0	35	0.010606	65/65
Serial	p_value1 (delta_1)	100	98	2	0	0.699313	98/100
Serial	p_value2 (delta_2)	100	98	2	0	0.971699	98/100
LinearComplexity	single	100	100	0	0	0.739918	100/100

Authentication-tag stream

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
Frequency	single	100	99	1	0	0.514124	99/100
BlockFrequency	single	100	100	0	0	0.574903	100/100
CumulativeSums	forward	100	100	0	0	0.534146	100/100
CumulativeSums	reverse	100	98	2	0	0.595549	98/100
Runs	single	100	99	1	0	0.759756	99/100
LongestRun	single	100	100	0	0	0.595549	100/100
Rank	single	100	96	4	0	0.137282	96/100
FFT	single	100	100	0	0	0.289667	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 0 1	100	99	1	0	0.574903	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 1 1	100	99	1	0	0.162606	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 0 1	100	99	1	0	0.383827	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 1 1	100	99	1	0	0.983453	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 0 1	100	100	0	0	0.657933	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 1 1	100	99	1	0	0.262249	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 0 1	100	99	1	0	0.494392	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 1 1	100	99	1	0	0.010988	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	100	0	0	0.137282	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	99	1	0	0.071177	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	100	0	0	0.494392	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	99	1	0	0.010237	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	100	0	0	0.637119	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	99	1	0	0.883171	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	100	0	0	0.304126	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	100	0	0	0.350485	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	97	3	0	0.213309	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 0	100	97	3	0	0.779188	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	98	2	0	0.637119	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 0	100	99	1	0	0.883171	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	96	4	0	0.534146	96/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 0	100	99	1	0	0.851383	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	99	1	0	0.759756	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	97	3	0	0.657933	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 0	100	100	0	0	0.678686	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	100	0	0	0.224821	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	100	0	0	0.574903	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	98	2	0	0.096578	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	99	1	0	0.678686	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	99	1	0	0.678686	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 0 1	100	99	1	0	0.075719	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 0	100	98	2	0	0.719747	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 1	100	99	1	0	0.699313	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 0 1	100	98	2	0	0.574903	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 0	100	99	1	0	0.554420	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 1	100	99	1	0	0.366918	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 0 1	100	98	2	0	0.514124	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 0	100	100	0	0	0.816537	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 1	100	98	2	0	0.213309	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 0 1	100	99	1	0	0.719747	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 0	100	100	0	0	0.616305	100/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 1 1	100	100	0	0	0.514124	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 0 1	100	99	1	0	0.739918	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 1 1	100	98	2	0	0.401199	98/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	100	0	0	0.474986	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 0 1	100	99	1	0	0.304126	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 1 1	100	100	0	0	0.935716	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 0 1	100	100	0	0	0.224821	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 1 1	100	100	0	0	0.366918	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 0 1 1	100	100	0	0	0.574903	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 0 1	100	99	1	0	0.946308	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 1 1	100	100	0	0	0.062821	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1	100	100	0	0	0.494392	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1	100	99	1	0	0.637119	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1	100	99	1	0	0.474986	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1	100	99	1	0	0.759756	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1	100	99	1	0	0.987896	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1	100	100	0	0	0.474986	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1	100	100	0	0	0.002374	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1	100	100	0	0	0.657933	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1	100	98	2	0	0.739918	98/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1	100	100	0	0	0.350485	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1	100	99	1	0	0.023545	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1	100	100	0	0	0.249284	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 0 1	100	100	0	0	0.137282	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1	100	95	5	0	0.554420	95/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 0 1	100	99	1	0	0.924076	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1	100	100	0	0	0.115387	100/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 0 1 1	100	99	1	0	0.867692	99/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 1 1 1	100	99	1	0	0.262249	99/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 0 1 1	100	99	1	0	0.262249	99/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 1 1 1	100	100	0	0	0.616305	100/100
NonOverlappingTemplate	m=9 template=0 1 1 1 0 1 1 1	100	99	1	0	0.637119	99/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 1 1 1	100	99	1	0	0.554420	99/100
NonOverlappingTemplate	m=9 template=1 0 0 0 0 0 0 0	100	99	1	0	0.574903	99/100
NonOverlappingTemplate	m=9 template=1 0 0 0 1 0 0 0	100	99	1	0	0.319084	99/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 0 0 0	100	99	1	0	0.719747	99/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 1 0 0	100	100	0	0	0.437274	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 0 0 0	100	98	2	0	0.554420	98/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 1 0 0	100	99	1	0	0.153763	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 0 0	100	100	0	0	0.719747	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 0 0	100	99	1	0	0.145326	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0	100	98	2	0	0.534146	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0	100	100	0	0	0.911413	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0	100	100	0	0	0.145326	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 1 0	100	99	1	0	0.366918	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0	100	98	2	0	0.334538	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 1 0	100	100	0	0	0.090936	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 0 0	100	100	0	0	0.137282	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 1 0	100	100	0	0	0.401199	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 0 0	100	97	3	0	0.474986	97/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 1 0	100	97	3	0	0.401199	97/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 0 0 0	100	98	2	0	0.719747	98/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 1 0 0	100	98	2	0	0.851383	98/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 0 0 0	100	100	0	0	0.816537	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 1 0 0	100	97	3	0	0.816537	97/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 0 0	100	99	1	0	0.637119	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 1 0 0	100	99	1	0	0.071177	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	100	0	0	0.040108	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	99	1	0	0.991468	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 1 0	100	99	1	0	0.637119	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 0 0	100	100	0	0	0.987896	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 1 0	100	98	2	0	0.554420	98/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 1 0 0	100	99	1	0	0.699313	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 0 0	100	98	2	0	0.383827	98/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 1 0	100	100	0	0	0.048716	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 0 0	100	99	1	0	0.366918	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 1 0	100	99	1	0	0.657933	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 1 0 0	100	99	1	0	0.779188	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 0 0	100	99	1	0	0.779188	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 1 0	100	98	2	0	0.759756	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 1 0 0	100	100	0	0	0.935716	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 0 0	100	99	1	0	0.080519	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 1 0	100	99	1	0	0.202268	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 1 0 0	100	98	2	0	0.419021	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 0 0	100	98	2	0	0.514124	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 1 0	100	100	0	0	0.534146	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 1 0 0	100	99	1	0	0.162606	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 0 0	100	99	1	0	0.851383	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 1 0	100	100	0	0	0.834308	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 0 0	100	98	2	0	0.137282	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 1 0	100	100	0	0	0.595549	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 0 0	100	99	1	0	0.236810	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 1 0	100	98	2	0	0.759756	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 1 0 0	100	98	2	0	0.213309	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 0 0	100	97	3	0	0.759756	97/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 1 0	100	99	1	0	0.595549	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 0 0	100	99	1	0	0.383827	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 1 0	100	99	1	0	0.289667	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 0 0	100	99	1	0	0.202268	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 1 0	100	97	3	0	0.350485	97/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 1 0 0	100	100	0	0	0.816537	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 0 0	100	100	0	0	0.883171	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 1 0	100	100	0	0	0.213309	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 0 0	100	100	0	0	0.304126	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 1 0	100	100	0	0	0.739918	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 0 0	100	99	1	0	0.419021	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 1 0	100	100	0	0	0.213309	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0 0	100	97	3	0	0.739918	97/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 1 0	100	99	1	0	0.474986	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0 0	100	99	1	0	0.699313	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 1 0	100	100	0	0	0.181557	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0 0	100	99	1	0	0.350485	99/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0	100	100	0	0	0.051942	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0	100	100	0	0	0.366918	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0	100	97	3	0	0.816537	97/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 0	100	100	0	0	0.129620	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 0	100	99	1	0	0.554420	99/100
OverlappingTemplate	single	100	98	2	0	0.108791	98/100
Universal	single	100	99	1	0	0.616305	99/100
ApproximateEntropy	single	100	98	2	0	0.924076	98/100
RandomExcursions	state=-4	62	60	2	38	0.534146	60/62
RandomExcursions	state=-3	62	57	5	38	0.028181	57/62
RandomExcursions	state=-2	62	62	0	38	0.706149	62/62
RandomExcursions	state=-1	62	62	0	38	0.834308	62/62
RandomExcursions	state=1	62	61	1	38	0.275709	61/62
RandomExcursions	state=2	62	62	0	38	0.671779	62/62
RandomExcursions	state=3	62	62	0	38	0.407091	62/62
RandomExcursions	state=4	62	62	0	38	0.964295	62/62
RandomExcursionsVariant	state=-9	62	61	1	38	0.671779	61/62
RandomExcursionsVariant	state=-8	62	61	1	38	0.862344	61/62
RandomExcursionsVariant	state=-7	62	60	2	38	0.834308	60/62
RandomExcursionsVariant	state=-6	62	61	1	38	0.739918	61/62
RandomExcursionsVariant	state=-5	62	61	1	38	0.253551	61/62
RandomExcursionsVariant	state=-4	62	62	0	38	0.534146	62/62
RandomExcursionsVariant	state=-3	62	60	2	38	0.110952	60/62
RandomExcursionsVariant	state=-2	62	60	2	38	0.500934	60/62
RandomExcursionsVariant	state=-1	62	60	2	38	0.054199	60/62
RandomExcursionsVariant	state=1	62	62	0	38	0.888137	62/62
RandomExcursionsVariant	state=2	62	62	0	38	0.637119	62/62
RandomExcursionsVariant	state=3	62	62	0	38	0.054199	62/62
RandomExcursionsVariant	state=4	62	62	0	38	0.437274	62/62
RandomExcursionsVariant	state=5	62	62	0	38	0.350485	62/62
RandomExcursionsVariant	state=6	62	62	0	38	0.437274	62/62
RandomExcursionsVariant	state=7	62	62	0	38	0.706149	62/62
RandomExcursionsVariant	state=8	62	62	0	38	0.602458	62/62
RandomExcursionsVariant	state=9	62	62	0	38	0.991468	62/62
Serial	p_value1 (delta_1)	100	97	3	0	0.978072	97/100
Serial	p_value2 (delta_2)	100	100	0	0	0.657933	100/100
LinearComplexity	single	100	99	1	0	0.699313	99/100

image_output_step2

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
Frequency	single	100	0	100	0	0.000000	0/100
BlockFrequency	single	100	0	100	0	0.000000	0/100
CumulativeSums	forward	100	0	100	0	0.000000	0/100
CumulativeSums	reverse	100	0	100	0	0.000000	0/100
Runs	single	100	0	100	0	0.000000	0/100
LongestRun	single	100	0	100	0	0.000000	0/100
Rank	single	100	0	100	0	0.000000	0/100
FFT	single	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 0 1	100	8	92	0	0.000000	8/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 1 1	100	8	92	0	0.000000	8/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 0 1	100	11	89	0	0.000000	11/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 0 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 1 1	100	18	82	0	0.000000	18/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 0 1	100	15	85	0	0.000000	15/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	7	93	0	0.000000	7/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	9	91	0	0.000000	9/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	29	71	0	0.000000	29/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	10	90	0	0.000000	10/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	44	56	0	0.000000	44/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	14	86	0	0.000000	14/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	2	98	0	0.000000	2/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 0	100	3	97	0	0.000000	3/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 0	100	11	89	0	0.000000	11/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 0	100	36	64	0	0.000000	36/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	33	67	0	0.000000	33/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 0	100	29	71	0	0.000000	29/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	17	83	0	0.000000	17/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 0	100	14	86	0	0.000000	14/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	27	73	0	0.000000	27/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 0 1	100	5	95	0	0.000000	5/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 1	100	48	52	0	0.000000	48/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 0 1	100	11	89	0	0.000000	11/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 0	100	24	76	0	0.000000	24/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 1	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 0 1	100	5	95	0	0.000000	5/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 1	100	8	92	0	0.000000	8/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 0 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 0	100	20	80	0	0.000000	20/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 0 1	100	23	77	0	0.000000	23/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 1 1	100	45	55	0	0.000000	45/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 0 1	100	2	98	0	0.000000	2/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 1 1	100	2	98	0	0.000000	2/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 0 1	100	5	95	0	0.000000	5/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 1 1	100	9	91	0	0.000000	9/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 0 1 1	100	10	90	0	0.000000	10/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 0 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1 1	100	52	48	0	0.000000	52/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1 1	100	26	74	0	0.000000	26/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1 1	100	25	75	0	0.000000	25/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1 1	100	21	79	0	0.000000	21/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1 1	100	17	83	0	0.000000	17/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1 1	100	19	81	0	0.000000	19/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1 1	100	25	75	0	0.000000	25/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 0 1 1	100	2	98	0	0.000000	2/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1 1	100	4	96	0	0.000000	4/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 0 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 0 1 1 1	100	21	79	0	0.000000	21/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 1 1 1 1	100	26	74	0	0.000000	26/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 0 1 1 1	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 1 1 0 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 0 0 0 0 0 0 0 0	100	8	92	0	0.000000	8/100
NonOverlappingTemplate	m=9 template=1 0 0 0 1 0 0 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 0 0 0 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 1 0 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 0 0 0 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 1 0 0 0	100	50	50	0	0.000000	50/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 0 0 0	100	8	92	0	0.000000	8/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 0 0 0	100	15	85	0	0.000000	15/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 0 0 0	100	4	96	0	0.000000	4/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 1 0 0	100	18	82	0	0.000000	18/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0 0	100	36	64	0	0.000000	36/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 1 0 0	100	7	93	0	0.000000	7/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0 0	100	9	91	0	0.000000	9/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 1 0 0	100	6	94	0	0.000000	6/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 0 0 0	100	19	81	0	0.000000	19/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 1 0 0	100	45	55	0	0.000000	45/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 0 0 0	100	55	45	0	0.000000	55/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 1 0 0	100	0	100	0	0.000000	0/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 0 0 0	100	38	62	0	0.000000	38/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 1 0 0	100	31	69	0	0.000000	31/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 0 0 0	100	3	97	0	0.000000	3/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 1 0 0	100	13	87	0	0.000000	13/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 0 0	100	11	89	0	0.000000	11/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 1 0	100	13	87	0	0.000000	13/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 1 0 0	100	8	92	0	0.000000	8/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	7	93	0	0.000000	7/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 0 0	100	16	84	0	0.000000	16/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 1 0 0	100	12	88	0	0.000000	12/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 0 0	100	36	64	0	0.000000	36/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 1 0	100	13	87	0	0.000000	13/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 0 0	100	6	94	0	0.000000	6/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 1 0 0	100	32	68	0	0.000000	32/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 0 0	100	35	65	0	0.000000	35/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 1 0 0	100	2	98	0	0.000000	2/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 0 0	100	31	69	0	0.000000	31/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 1 0	100	14	86	0	0.000000	14/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 1 0 0	100	16	84	0	0.000000	16/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 0 0	100	9	91	0	0.000000	9/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 1 0	100	15	85	0	0.000000	15/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 1 0 0	100	12	88	0	0.000000	12/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 0 0	100	5	95	0	0.000000	5/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 0 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 1 0	100	43	57	0	0.000000	43/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 0 0	100	4	96	0	0.000000	4/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 1 0	100	37	63	0	0.000000	37/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 1 0 0	100	25	75	0	0.000000	25/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 0 0	100	4	96	0	0.000000	4/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 1 0	100	45	55	0	0.000000	45/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 0 0	100	43	57	0	0.000000	43/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 0 0	100	6	94	0	0.000000	6/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 1 0	100	10	90	0	0.000000	10/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 1 0 0	100	11	89	0	0.000000	11/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 0 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 1 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 1 0	100	12	88	0	0.000000	12/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 1 0	100	32	68	0	0.000000	32/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 1 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0 0	100	0	100	0	0.000000	0/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 0	100	0	100	0	0.000000	0/100
OverlappingTemplate	single	100	0	100	0	0.000000	0/100
Universal	single	100	0	100	0	0.000000	0/100
ApproximateEntropy	single	100	0	100	0	0.000000	0/100
RandomExcursions	state=-4	0	0	0	100	----	-----
RandomExcursions	state=-3	0	0	0	100	----	-----
RandomExcursions	state=-2	0	0	0	100	----	-----
RandomExcursions	state=-1	0	0	0	100	----	-----
RandomExcursions	state=1	0	0	0	100	----	-----
RandomExcursions	state=2	0	0	0	100	----	-----
RandomExcursions	state=3	0	0	0	100	----	-----
RandomExcursions	state=4	0	0	0	100	----	-----
RandomExcursionsVariant	state=-9	0	0	0	100	----	-----
RandomExcursionsVariant	state=-8	0	0	0	100	----	-----
RandomExcursionsVariant	state=-7	0	0	0	100	----	-----
RandomExcursionsVariant	state=-6	0	0	0	100	----	-----
RandomExcursionsVariant	state=-5	0	0	0	100	----	-----
RandomExcursionsVariant	state=-4	0	0	0	100	----	-----
RandomExcursionsVariant	state=-3	0	0	0	100	----	-----
RandomExcursionsVariant	state=-2	0	0	0	100	----	-----
RandomExcursionsVariant	state=-1	0	0	0	100	----	-----
RandomExcursionsVariant	state=1	0	0	0	100	----	-----
RandomExcursionsVariant	state=2	0	0	0	100	----	-----
RandomExcursionsVariant	state=3	0	0	0	100	----	-----
RandomExcursionsVariant	state=4	0	0	0	100	----	-----
RandomExcursionsVariant	state=5	0	0	0	100	----	-----
RandomExcursionsVariant	state=6	0	0	0	100	----	-----
RandomExcursionsVariant	state=7	0	0	0	100	----	-----
RandomExcursionsVariant	state=8	0	0	0	100	----	-----
RandomExcursionsVariant	state=9	0	0	0	100	----	-----
Serial	p_value1 (delta_1)	100	0	100	0	0.000000	0/100
Serial	p_value2 (delta_2)	100	0	100	0	0.000000	0/100
LinearComplexity	single	100	97	3	0	0.616305	97/100

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NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
Frequency	single	100	0	100	0	0.000000	0/100
BlockFrequency	single	100	0	100	0	0.000000	0/100
CumulativeSums	forward	100	0	100	0	0.000000	0/100
CumulativeSums	reverse	100	0	100	0	0.000000	0/100
Runs	single	100	0	100	0	0.000000	0/100
LongestRun	single	100	0	100	0	0.000000	0/100
Rank	single	100	0	100	0	0.000000	0/100
FFT	single	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 0 1	100	4	96	0	0.000000	4/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 1 1	100	9	91	0	0.000000	9/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 0 1	100	11	89	0	0.000000	11/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 0 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 1 1	100	18	82	0	0.000000	18/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 0 1	100	16	84	0	0.000000	16/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	10	90	0	0.000000	10/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	12	88	0	0.000000	12/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	36	64	0	0.000000	36/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	2	98	0	0.000000	2/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	36	64	0	0.000000	36/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	13	87	0	0.000000	13/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 0	100	4	96	0	0.000000	4/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	2	98	0	0.000000	2/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 0	100	17	83	0	0.000000	17/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	5	95	0	0.000000	5/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 0	100	40	60	0	0.000000	40/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	26	74	0	0.000000	26/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 0	100	37	63	0	0.000000	37/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	24	76	0	0.000000	24/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 0	100	14	86	0	0.000000	14/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	26	74	0	0.000000	26/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 0 1	100	5	95	0	0.000000	5/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 0	100	2	98	0	0.000000	2/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 1	100	46	54	0	0.000000	46/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 0 1	100	12	88	0	0.000000	12/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 0	100	19	81	0	0.000000	19/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 0 1	100	8	92	0	0.000000	8/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 1	100	10	90	0	0.000000	10/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 0 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 0	100	21	79	0	0.000000	21/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 0 1	100	25	75	0	0.000000	25/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 1 1	100	44	56	0	0.000000	44/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 0 1	100	2	98	0	0.000000	2/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 1 1	100	11	89	0	0.000000	11/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 0 1	100	5	95	0	0.000000	5/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 1 1	100	18	82	0	0.000000	18/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 0 1 1	100	17	83	0	0.000000	17/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 0 1	100	2	98	0	0.000000	2/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1 1	100	59	41	0	0.000000	59/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1 1	100	26	74	0	0.000000	26/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1 1	100	24	76	0	0.000000	24/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1 1	100	19	81	0	0.000000	19/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1 1	100	23	77	0	0.000000	23/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1 1	100	26	74	0	0.000000	26/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1 1	100	23	77	0	0.000000	23/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 0 1 1	100	6	94	0	0.000000	6/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1 1	100	6	94	0	0.000000	6/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 0 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 0 1 1 1	100	24	76	0	0.000000	24/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 1 1 1 1	100	27	73	0	0.000000	27/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 0 1 1 1	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 1 1 0 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 1 1 1 1	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 0 0 0 0 0 0 0 0	100	4	96	0	0.000000	4/100
NonOverlappingTemplate	m=9 template=1 0 0 0 1 0 0 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 0 0 0 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 1 0 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 0 0 0 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 1 0 0 0	100	51	49	0	0.000000	51/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 0 0 0	100	5	95	0	0.000000	5/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 1 0 0	100	13	87	0	0.000000	13/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 0 0 0	100	6	94	0	0.000000	6/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 1 0 0	100	6	94	0	0.000000	6/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0 0	100	32	68	0	0.000000	32/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 1 0 0	100	8	92	0	0.000000	8/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0 0	100	14	86	0	0.000000	14/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 1 0 0	100	7	93	0	0.000000	7/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 0 0 0	100	17	83	0	0.000000	17/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 1 0 0	100	35	65	0	0.000000	35/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 0 0 0	100	57	43	0	0.000000	57/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 1 0 0	100	0	100	0	0.000000	0/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 0 0 0	100	40	60	0	0.000000	40/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 1 0 0	100	33	67	0	0.000000	33/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 0 0 0	100	15	85	0	0.000000	15/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 1 0 0	100	10	90	0	0.000000	10/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 0 0	100	12	88	0	0.000000	12/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 1 0	100	13	87	0	0.000000	13/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 1 0 0	100	5	95	0	0.000000	5/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	10	90	0	0.000000	10/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 0 0	100	18	82	0	0.000000	18/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 1 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 1 0 0	100	15	85	0	0.000000	15/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 0 0	100	33	67	0	0.000000	33/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 1 0	100	16	84	0	0.000000	16/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 0 0	100	7	93	0	0.000000	7/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 1 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 1 0 0	100	32	68	0	0.000000	32/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 0 0	100	39	61	0	0.000000	39/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 1 0 0	100	4	96	0	0.000000	4/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 0 0	100	30	70	0	0.000000	30/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 1 0	100	14	86	0	0.000000	14/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 1 0 0	100	16	84	0	0.000000	16/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 0 0	100	10	90	0	0.000000	10/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 1 0	100	16	84	0	0.000000	16/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 1 0 0	100	13	87	0	0.000000	13/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 0 0	100	3	97	0	0.000000	3/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 0 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 1 0	100	40	60	0	0.000000	40/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 0 0	100	5	95	0	0.000000	5/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 1 0	100	42	58	0	0.000000	42/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 1 0 0	100	29	71	0	0.000000	29/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 0 0	100	3	97	0	0.000000	3/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 1 0	100	49	51	0	0.000000	49/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 0 0	100	42	58	0	0.000000	42/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 1 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 0 0	100	5	95	0	0.000000	5/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 1 0	100	9	91	0	0.000000	9/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 1 0 0	100	15	85	0	0.000000	15/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 0 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 1 0	100	3	97	0	0.000000	3/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 1 0	100	16	84	0	0.000000	16/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 1 0	100	32	68	0	0.000000	32/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 1 0	100	4	96	0	0.000000	4/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 1 0	100	2	98	0	0.000000	2/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0 0	100	1	99	0	0.000000	1/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0	100	1	99	0	0.000000	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 0	100	0	100	0	0.000000	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 0	100	0	100	0	0.000000	0/100
OverlappingTemplate	single	100	0	100	0	0.000000	0/100
Universal	single	100	0	100	0	0.000000	0/100
ApproximateEntropy	single	100	0	100	0	0.000000	0/100
RandomExcursions	state=-4	0	0	0	100	----	-----
RandomExcursions	state=-3	0	0	0	100	----	-----
RandomExcursions	state=-2	0	0	0	100	----	-----
RandomExcursions	state=-1	0	0	0	100	----	-----
RandomExcursions	state=1	0	0	0	100	----	-----
RandomExcursions	state=2	0	0	0	100	----	-----
RandomExcursions	state=3	0	0	0	100	----	-----
RandomExcursions	state=4	0	0	0	100	----	-----
RandomExcursionsVariant	state=-9	0	0	0	100	----	-----
RandomExcursionsVariant	state=-8	0	0	0	100	----	-----
RandomExcursionsVariant	state=-7	0	0	0	100	----	-----
RandomExcursionsVariant	state=-6	0	0	0	100	----	-----
RandomExcursionsVariant	state=-5	0	0	0	100	----	-----
RandomExcursionsVariant	state=-4	0	0	0	100	----	-----
RandomExcursionsVariant	state=-3	0	0	0	100	----	-----
RandomExcursionsVariant	state=-2	0	0	0	100	----	-----
RandomExcursionsVariant	state=-1	0	0	0	100	----	-----
RandomExcursionsVariant	state=1	0	0	0	100	----	-----
RandomExcursionsVariant	state=2	0	0	0	100	----	-----
RandomExcursionsVariant	state=3	0	0	0	100	----	-----
RandomExcursionsVariant	state=4	0	0	0	100	----	-----
RandomExcursionsVariant	state=5	0	0	0	100	----	-----
RandomExcursionsVariant	state=6	0	0	0	100	----	-----
RandomExcursionsVariant	state=7	0	0	0	100	----	-----
RandomExcursionsVariant	state=8	0	0	0	100	----	-----
RandomExcursionsVariant	state=9	0	0	0	100	----	-----
Serial	p_value1 (delta_1)	100	0	100	0	0.000000	0/100
Serial	p_value2 (delta_2)	100	0	100	0	0.000000	0/100
LinearComplexity	single	100	99	1	0	0.419021	99/100

Individual p-values and official reports

- results.csv contains one row for every stream and component with the individual first-level p-value, status, pass proportion, uniformity p-value, alpha, bit length, NIST version, and raw input stream hash. N/A p-values are blank and explicitly labeled not applicable.
- per_test_summary.csv contains the complete per-test/component summary.
- raw_reports/ preserves complete STS finalAnalysisReport.txt, per-test stats.txt, master results.txt, and official component partition files for all six categories.
- The three professor-requested report aliases are cryptographic_components_finalAnalysisReport.txt, image_output_step2_finalAnalysisReport.txt, and image_output_final_marked_finalAnalysisReport.txt.
- raw_streams/ contains the exact packed binary NIST inputs. stream_manifest.csv records each stream's byte offset, size, hash, key/ImageID digests and source details.

Provenance and reproducibility

See provenance.txt, stream_manifest.csv, component_contexts.csv, and nist_sts_source_manifest.csv. They record source and executable hashes, runtime versions, platform, exact command and settings, input image hashes, per-stream hashes, and digest-only test credential identifiers. No test key or ImageID bytes are written.

Limitations

- NIST SP 800-22 is not a security proof and does not replace cryptanalysis.
- Image-output streams preserve image-related information by design; NIST results on those streams are diagnostics, not proof of security or insecurity.
- NPCR/UACI and NIST outcomes do not prove cryptographic security.
- The evaluation uses six public UCT images because the eight CelebA-HQ source files were unavailable. Authenticated-TPE is implemented with a fixed 32x32 block size; authenticated 8x8 and 16x16 results are therefore not reported.
- The component fixtures are deterministic and public for reproducibility. NIST testing does not assess entropy quality, key secrecy, unpredictability, formal security, or independent cryptanalysis.

References

1. NIST, SP 800-22 Rev. 1a. NIST notes statistical testing cannot substitute for cryptanalysis.
2. NIST, SP 800-22 documentation and software, including the NIST STS download and software revision history.